FOUNDATION OF DATA CENTRIC AI

A GENERAL VIEW

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Model Centric AI

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Data Centric AI

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Data quality errors

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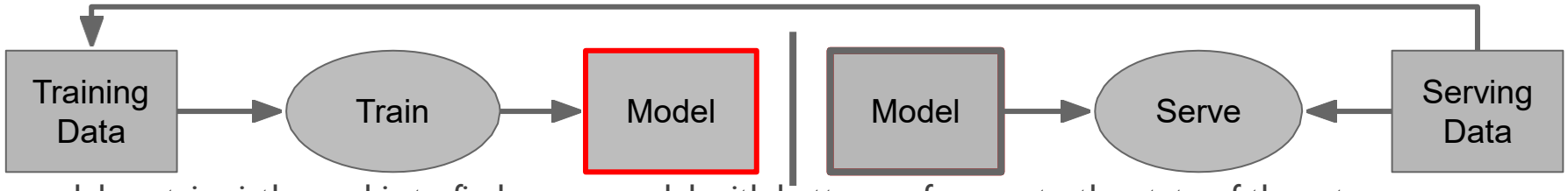
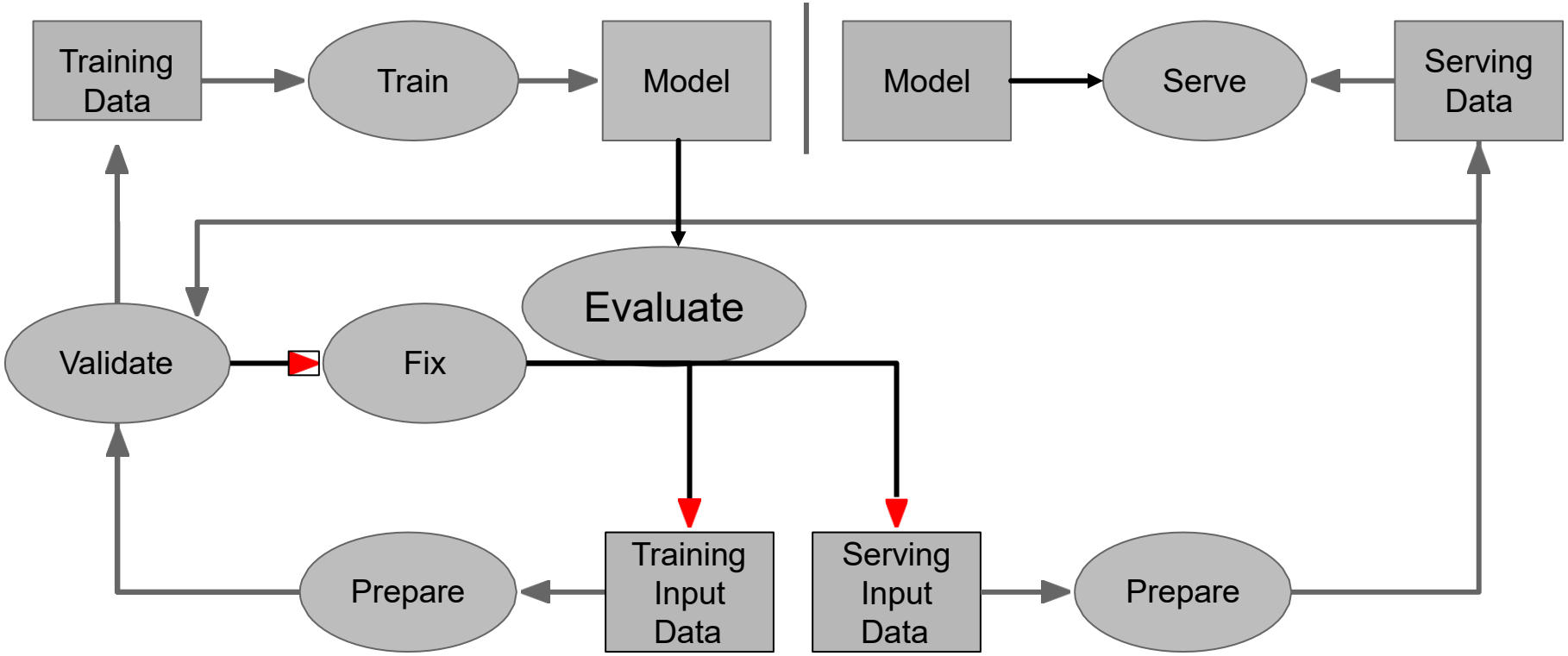
Data quality in AI

A TYPICAL PIPELINE

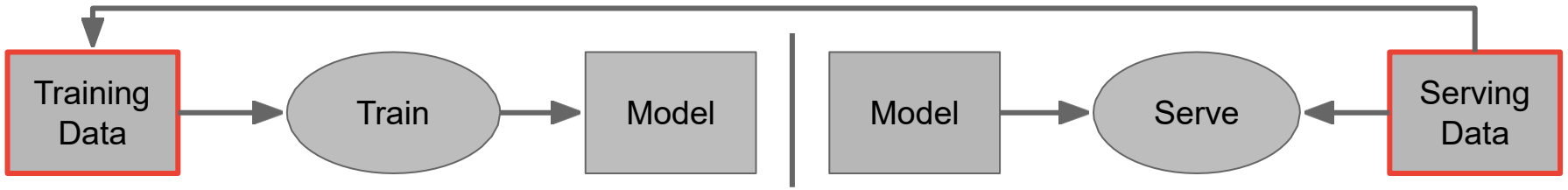


- In the model centric ai: the goal is to find a new model with better preference to the state of the art
- Standard benchmark:
  - COCO (Common Objects in Context): [1]
  - OpenML-CTR[2]
  - Humanity's Last Exam (HLE)[3]
  - GSM8K[4]
- Leaderboard
  - OpenML Benchmarking [5]
  - Open LLM Leaderboard (Hugging Face) [6]
  - VLM Leaderboard (Vision-Language Models) [7]

THE REAL PIPELINE



THE DATA POINT-OF-VIEW

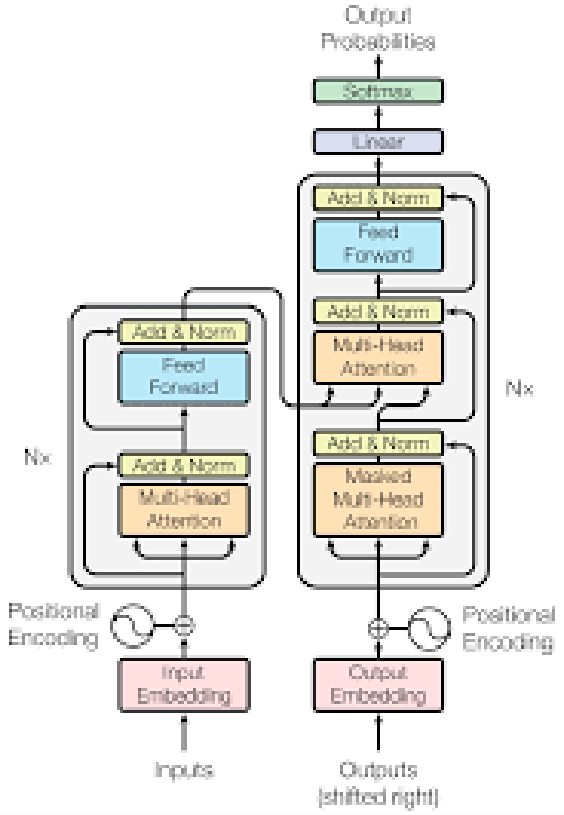


- What data-management issues arise when deploying ML in production?

- Having the right data is crucial for model quality.
- Preparing data for an ML pipeline requires effort and care.
- Invalid data can cause outages in production ⇒ data monitoring, validation, and fixing are essential.

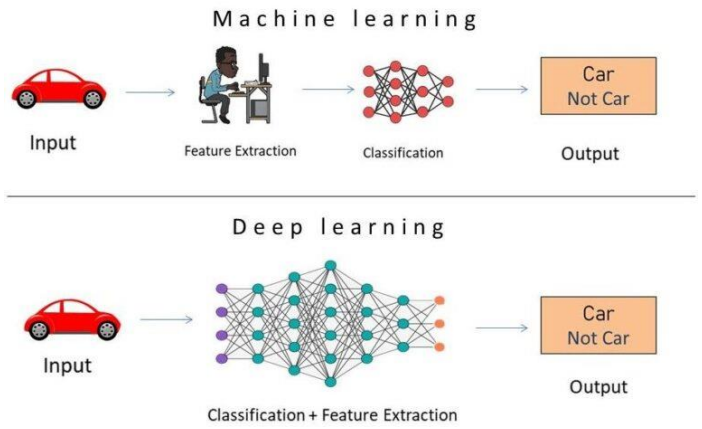
MODEL-CENTRIC APPROACH

- Dataset considered immutable
- Improvements focus on:
  - Model architectures
    - Attention is all that you need. "A new simple network architecture, the **Transformer**, based solely on **attention mechanisms**, dispensing with recurrence and convolutions entirely." [8]
  - Hyperparameter tuning
    - HyperBand. «Hyperband is a bandit-based strategy that allocates resources to a set of random configurations, using a principled early-stopping strategy to quickly eliminate poor-performing ones.»[9]
  - Optimization strategies
    - Adam [10]
  - Regularization techniques
    - DropOut[11]



WHY MODEL-CENTRIC WORKED (FOR 15 YEARS)

- Bigger Datasets
  - Availability of massive supervised datasets enabled deep learning training
  - Standardized benchmarks allowed reproducible comparison across models
  - ImageNet introduced learning at unprecedented visual scale (≈14M images) [12]
- New architecture (Deep Learning)
  - Deep neural networks demonstrated superior representation learning
  - End-to-end feature learning replaced manual feature engineering
  - Performance gains came primarily from architectural innovation
- New Hardware
  - GPUs enabled efficient large-scale matrix computation
  - Training time reduced from weeks to days
  - Larger models became experimentally feasible



THE BENCHMARK CULTURE IN MACHINE LEARNING

- Typical experimental workflow:
  - A fixed public dataset
  - A fixed evaluation metric
  - Direct comparison across models
  - Public leaderboards tracking performance
- Leaderboard race → Research optimization is increasingly focused on:
  - Achieving state-of-the-art (SOTA)
  - Small numerical improvements
    - +0.3% improvement (what is means? is that innovation?)
- Positive effects:
  - Standardized evaluation
  - Rapid technological progress
  - Reproducibility within benchmarks
- Unintended consequences:
  - Incremental improvements dominate research
  - Overfitting to benchmark datasets
  - Limited attention to data quality
  - Real-world generalization often ignored

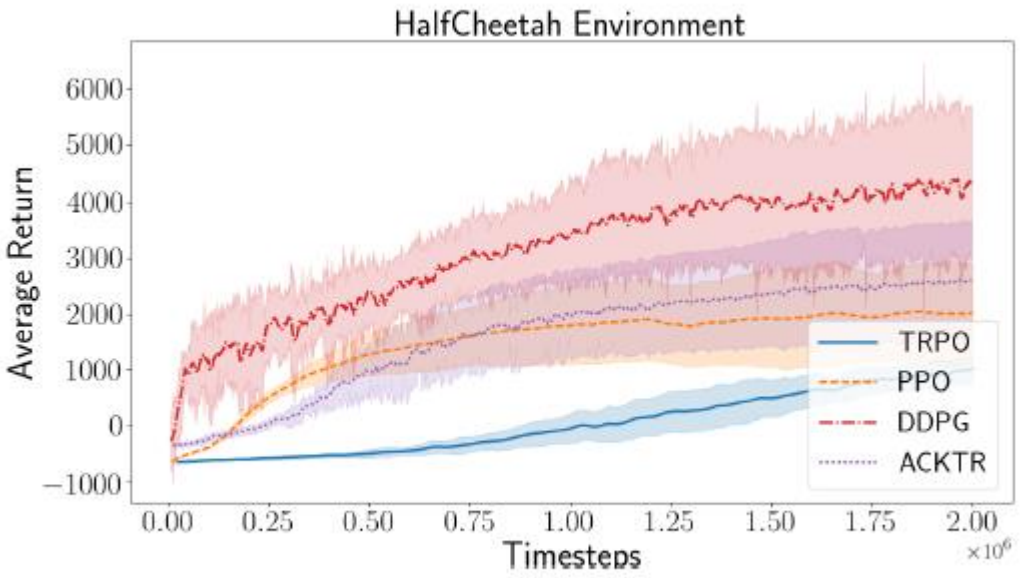
THE CASE OF IMAGENET

- "Do ImageNet Classifiers Generalize to ImageNet?" di Recht et al. (2019)[13]
- a new ImageNet test set was created from scratch, following the original 2009 data collection protocol.
  - the labels, source images, and selection processes were identical to the original to test "natural" generalization.

- Significant Accuracy Drop:
  - Every tested model saw a performance decline of 3% to 15% on the new data, despite it being virtually identical to the original.
- Linear Correlation:
  - A strong linear relationship exists: models that perform better on the original set also perform better on the new one, but the "generalization gap" remains.
- Dataset Fragility:
  - The results suggest that progress in Vision might be partially driven by models adapting to the specific biases of the ImageNet benchmark.

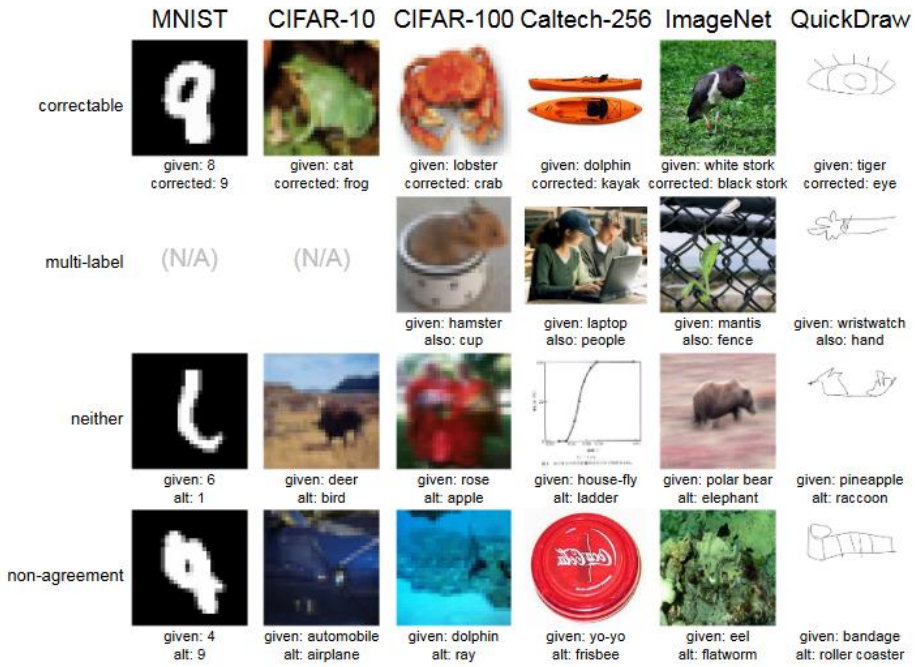
THE SATURATION PROBLEM

- Rapid improvements during early deep learning years → Gradual performance plateaus major benchmarks → Remaining gains increasingly marginal
- Improvements smaller than 1% accuracy
- Are these improvements meaningful outside the benchmark?
- Reproducibility crisis
  - Empirical findings showed that results often depend on:
    - Random initialization
    - Training instability
    - Undocumented preprocessing steps
    - Hardware and software differences
  - Same method → different outcomes across runs or labs.[14]
- Hyperparameter lottery
- Performance may depend more on tuning than innovation:
  - Learning rate schedules
  - Batch size
  - Initialization
  - Training duration
- Reported improvements may reflect optimization effort rather than scientific advancement.



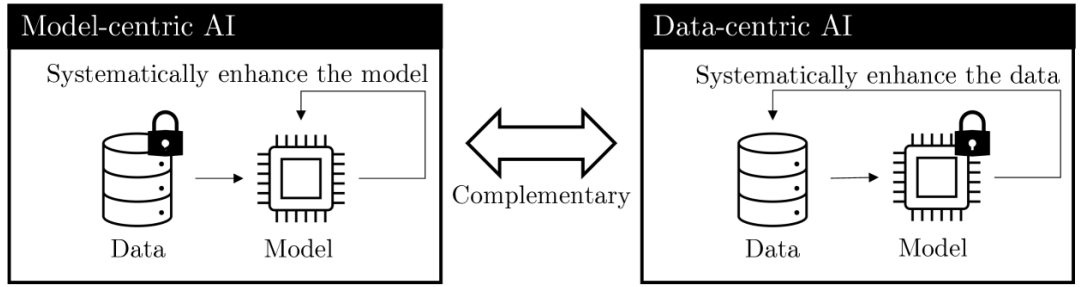
WHAT ABOUT DATA

- Machine learning experiments implicitly assume datasets are fixed and reliable
- BUT, datasets are the result of many design choices:
  - Data collection procedures
  - Sampling strategies
  - Annotation protocols
  - Filtering and preprocessing decisions
  - Datasets embed human decisions:
    - Annotation guidelines, Annotator expertise Cultural and contextual assumptions, Economic constraints during labeling
- Dataset construction introduces systematic bias.
- Are labels clean?
  - Annotator disagreement, Ambiguous examples Conceptual uncertainty Evolving definitions Empirical evidence:
  - Large benchmark datasets contain non-negligible label errors. [15]
- Are they representative? [16]
  - Early facial recognition datasets (e.g., LFW) were predominantly composed of lighter-skinned male faces, often sourced from Western academic environments.
  - Key Finding: Commercial AI systems showed massive error rates for darker-skinned females (up to 34.7%) compared to lighter-skinned males (0.8%).



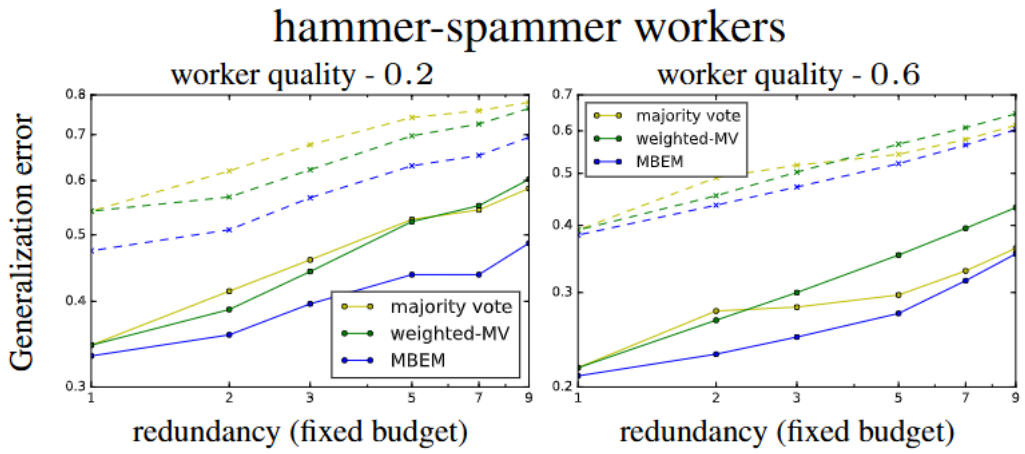
DATA CENTRIC AI

- From model optimization to dataset optimization
- Machine learning performance depends as much on data quality as on model design.
- The term Data-Centric AI was popularized by: Andrew Ng [17]
- Traditional assumption:
  - Dataset fixed → optimize the model
- Data-centric perspective:
  - Model relatively stable → iteratively improve the dataset
- Examples of dataset improvement:
  - Label correction, Better sampling strategies, Noise detection, Dataset balancing Improved annotation guidelines
- Data is now treated a design variable, object of optimization as a controllable component of experimentation
- Dataset construction becomes part of the scientific method.



DATA CENTRIC AI

- Typical data centric topic:
- Error detection
  - Identify mislabeled or ambiguous samples[18]
- Data cleaning
  - Remove corrupted or inconsistent observations
  - Active clean [19]
- Re-labeling
  - Refine annotations using improved guidelines or expert review[20]
- Data augmentation
  - Increase robustness through controlled data transformations[21]
- Bias correction
  - Adjust sampling to improve representativeness [22]



WHEN USE A DATA CENTRIC APPROACH

- Small or Medium-Sized Datasets
  - Model capacity already exceeds available information
  - Noise and labeling errors dominate learning behavior
  - Overfitting risk increases
  - Improving data quality often produces larger gains than increasing model complexity
- Domain-Specific Machine Learning
  - Data distributions differ from public benchmarks
  - Pretrained models are only partially aligned
  - Expert annotation is required
- When Foundation Models Are Already Strong
  - Pretrained models provide strong baselines
  - Architecture improvements yield marginal gains
  - Performance differences depend heavily on input data quality

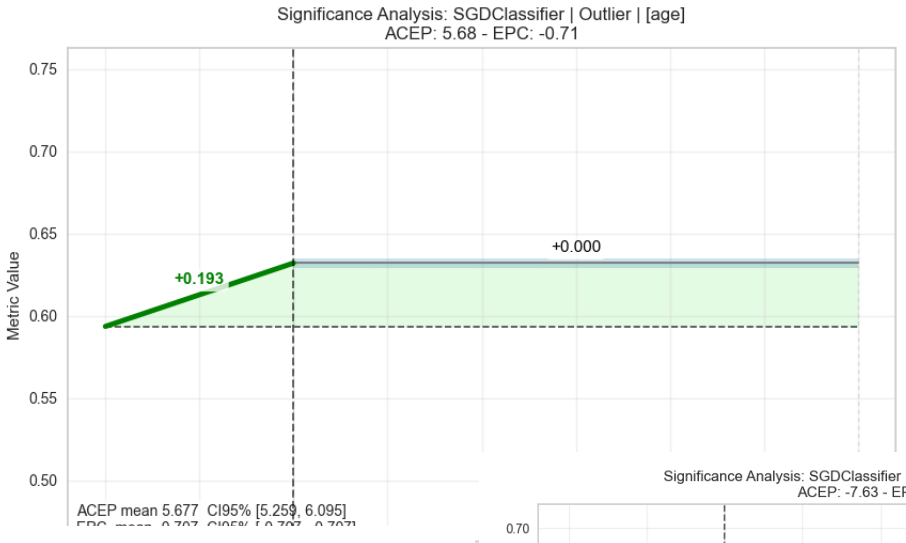
Model	Precision	Recall	F1-Score
Llama_3B	0.8421	**0.8491**	0.8440
Llama2_7B	**0.8491**	**0.8491**	**0.8491**
Llama3_8B	0.8290	0.7925	0.8036
Gemma_2B	0.8176	0.8302	0.8127
Gemma_7B	0.8190	0.8302	0.8211
RecurrentGemma_2B	0.8470	0.5283	0.5452
Mistral_7B	0.8032	0.7736	0.7840
GPT4o_mini	0.7684	0.7925	0.7711
ESG-BERT	0.8360	0.6792	0.7050

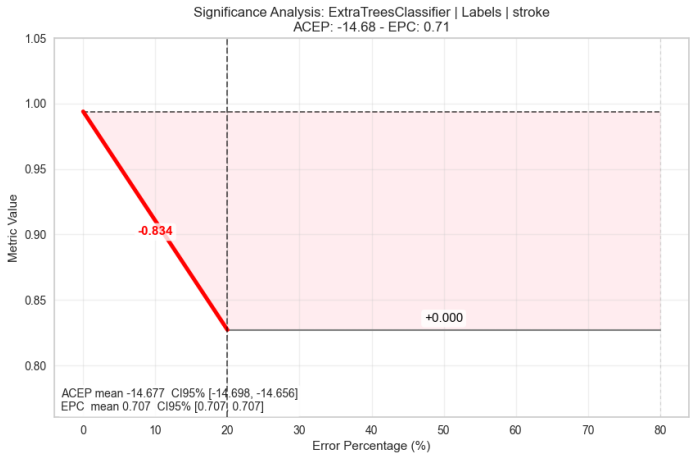
Table 4: Performance of the fine-tuned models (original and synthetic data).

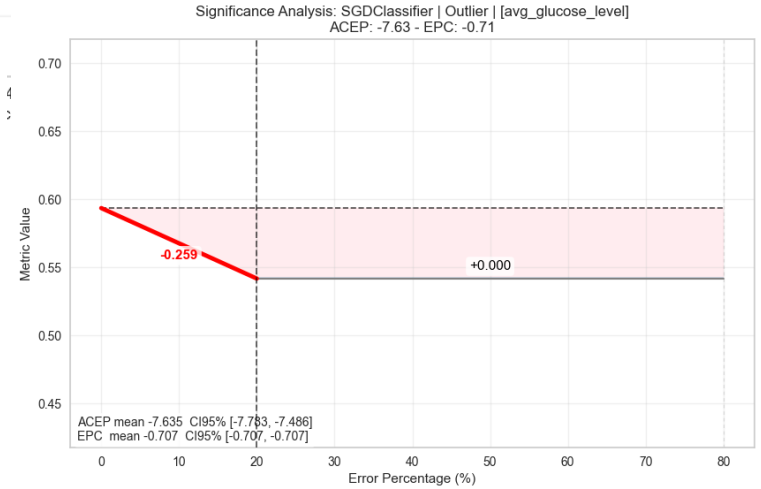
- LLM fine tuning[23]

TABULAR ML

- In tabular settings, model architectures are often mature and comparable in performance (e.g., tree ensembles, gradient boosting, shallow neural networks).
- Common data issues:
  - Missing values
  - Measurement errors
  - Selection bias
  - Temporal drift
- Impact often larger than model choice.



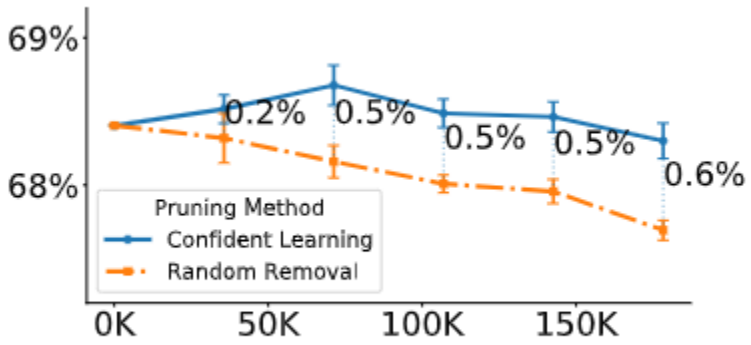




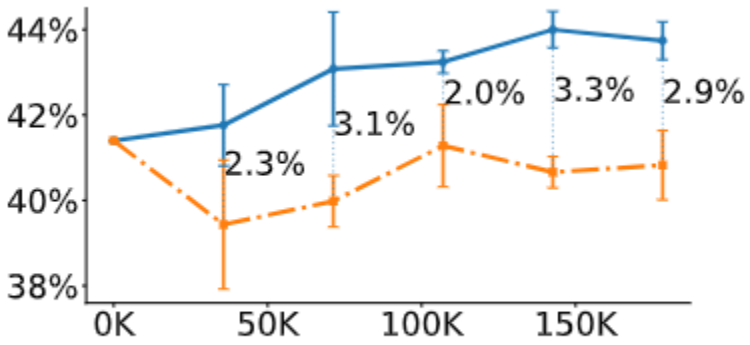
VISION

- Experimental observation:
- Remove ~0.5% label noise
- Significant accuracy improvement[24]

- Another example [25]



(a) Accuracy on the ILSVRC2012 validation set



(b) Accuracy on the top 20 noisiest classes

EXAMPLE NLP

- Ambiguity in Language Understanding
  - Ambiguous sentiment labels
  - Annotation disagreement
  - Dataset bias
- Human annotators frequently disagree even when:[26]
  - instructions are clear
  - annotators are trained
  - tasks are well-defined
- Crowd annotation studies show:
  - ambiguity is intrinsic to language
  - majority voting may remove meaningful variation

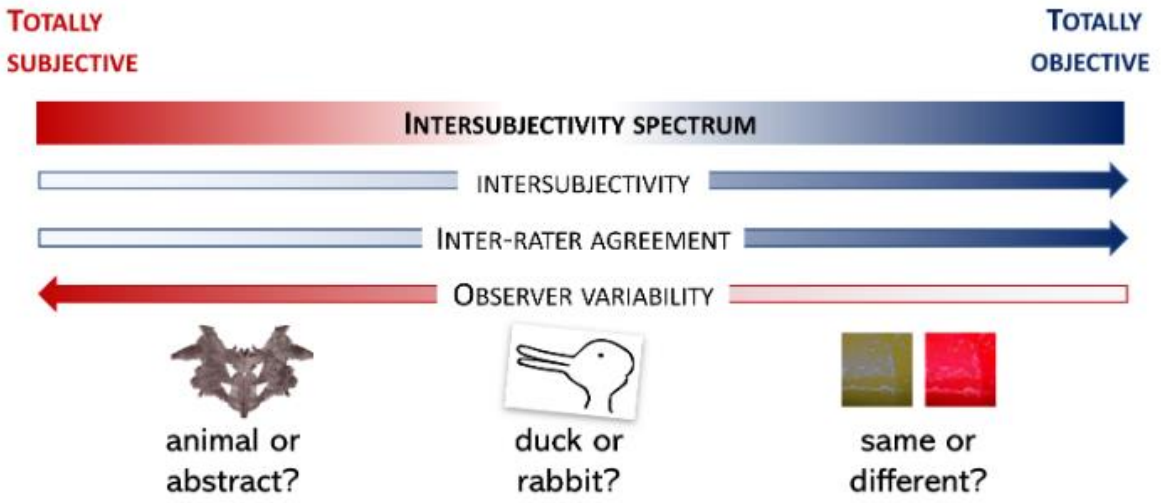
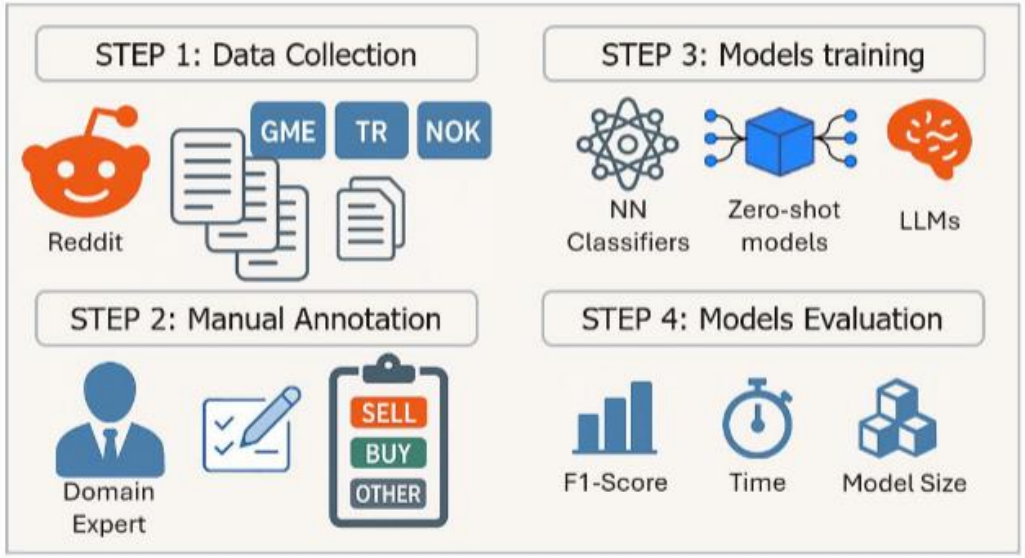


Fig. 2. The intersubjectivity spectrum and its relationship with inter-rater agreement.

FONDATION MODEL

- Recent advances introduced a new paradigm:
  - large pretrained models trained once
  - reused across many downstream tasks
  - adapted mainly through data rather than architecture redesign
- Pretrained Models as Universal Starting Points
- New Interaction Paradigms
  - Prompting
  - Fine-tuning
  - Instruction tuning / alignment
- Old paradigm:
  - New task → design new model
- New paradigm:
  - New task → curate better data for an existing model

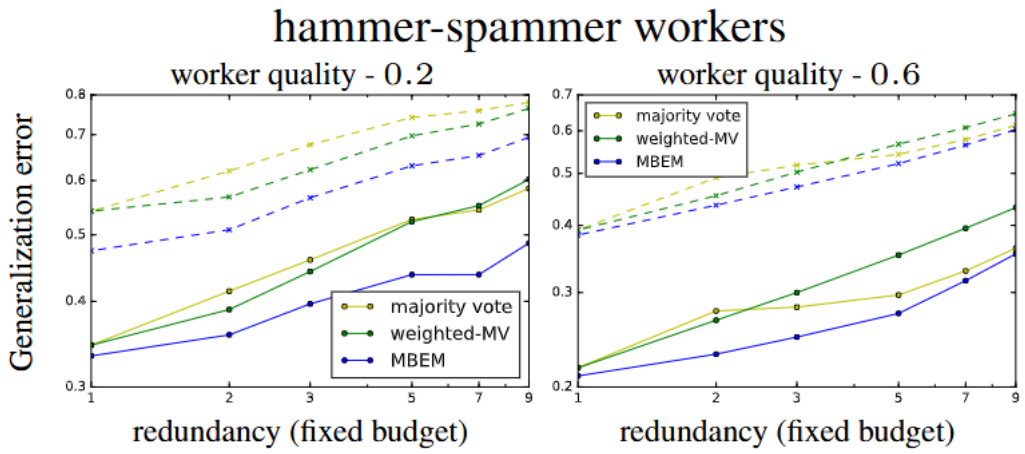
- Social Trading Action Detection (STAD),

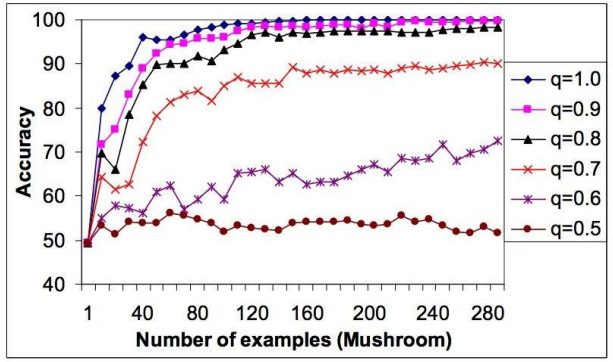


DATA SCALE

- Performance improves as we increase:
  - training data
  - model parameters
  - computational budget

- BUT:
  - improvements follow a power-law trend
  - each additional increase in data produces smaller gains
- doubling data does not double performance
- Bigger is better? [27]





DATA AS ADDED VALUE OF COMPANIES

- In the foundation model era, many organizations start from the same underlying models, yet achieve very different outcomes.
- The primary source of differentiation shifts from models to data.
- Modern AI development often relies on:
  - shared pretrained architectures
  - publicly available foundation models
  - similar optimization techniques
- Model architecture is no longer the main barrier to entry.
- Different Proprietary Data
- Organizations differ primarily in:
  - domain-specific datasets
  - curated internal corpora
  - user interaction data
  - high-quality annotations
  - feedback and operational data

- Different Performance Outcomes
- Even with identical base models:
- adaptation datasets differ
- fine-tuning signals differ
- evaluation distributions differ

MODEL CENTRIC VS DATA CENTRIC

Model-Centric	Data-Centric
Architecture innovation	Dataset improvement
Hyperparameters	Label quality
Loss engineering	Bias detection
Training tricks	Sampling strategy

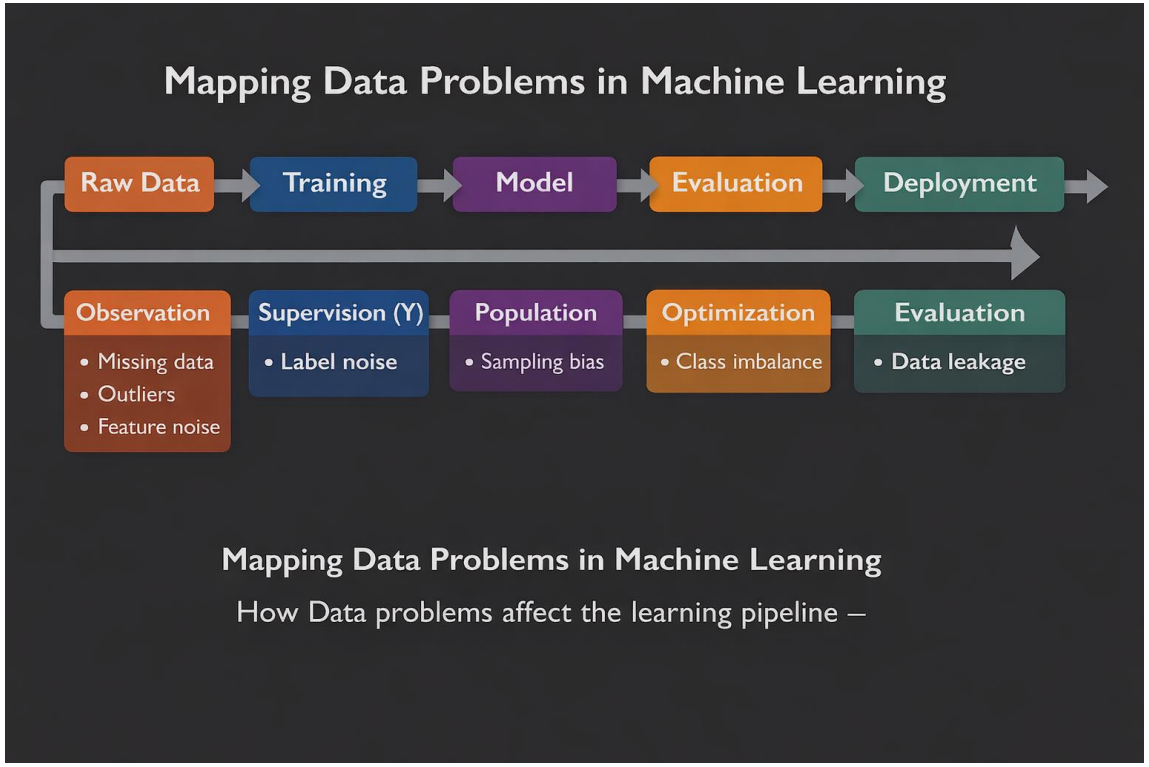
ERROR DATA TAXONOMY

A TAXONOMY OF DATA PROBLEMS IN ML/DL/GENAI

- Why Study Data Failures?
  - Data-centric AI requires understanding how and why datasets fail
  - Data issues are not homogeneous: different problems affect different stages of learning
  - Model performance degradation often originates from specific data failure modes
- Data problems are related but different from traditional data quality issues
- Bad data is not a single problem
  - missing values, wrong labels, bias, imbalance produce different effect on predictive task
- Data problems arise from different failure mechanisms:
  - different statistical causes
  - different effects on learning
  - different impacts on generalization

HOW DATA PROBLEMS AFFECT MACHINE LEARNING

- Machine learning failures often originate from different types of data imperfections, each acting on a distinct component of the learning process.
  - Features noise
    - Outliers. extreme or anomalous observations
    - Missing data incomplete acquisition processes
    - Noisy data due to sensors or human input
  - Failure modes
    - Label Noise, Incorrect or ambiguous labels [25]
    - Sampling Bias, training data does not reflect the true population distribution.[28]
    - Class Imbalance. Rare classes receive insufficient learning signal,[29]
    - Data shift. Training and deployment distributions differ [30]
    - Data Leakage. Information from the test or future data contaminates training [31]



FEATURE NOISE

- Feature noise affects the observed inputs:

$$x_{observed} = x_{true} + \epsilon$$

- where:
  - x_{true} =latent real measurement
  - ϵ= measurement error or perturbation.
- Feature noise degrades the **quality of the signal**, not the supervision.

- Typical Sources
  - sensor errors
  - measurement inaccuracies
  - preprocessing artifacts
  - data integration pipelines
- Examples:
  - financial indicators with delayed updates
  - medical measurements variability
  - noisy text embeddings

OUTLIER

- Extreme observations inconsistent with the underlying distribution.[33]
  - faulty sensors
  - rare events
  - logging errors
  - distribution mixture effects
- Effect:
  - unstable decision boundaries
  - distorted statistical estimates.

- An outlier is a statistical deviation, not necessarily a mistake.
- Outliers may represent noise, rare events, or new phenomena.
- Outliers often correspond to:
  - heavy-tailed distributions
  - mixture distributions
  - regime changes
- $P_{data}(x) \neq$ *single Gaussian assumption*

OUTLIER AND DATA SHIFT

- Outliers are often treated as isolated anomalies.
- in evolving systems
- They may represent early signals of changes in the underlying data-generating process

- Let training data follow:

$$x \sim P_{train}(X)$$

- During deployment:

$$x \sim P_{deploy}(X)$$

- Distribution shift occurs when:

$$P_{train}(X) \neq P_{deploy}(X)$$

- Removing all outliers may remove early evidence of future data distributions.

MISSING

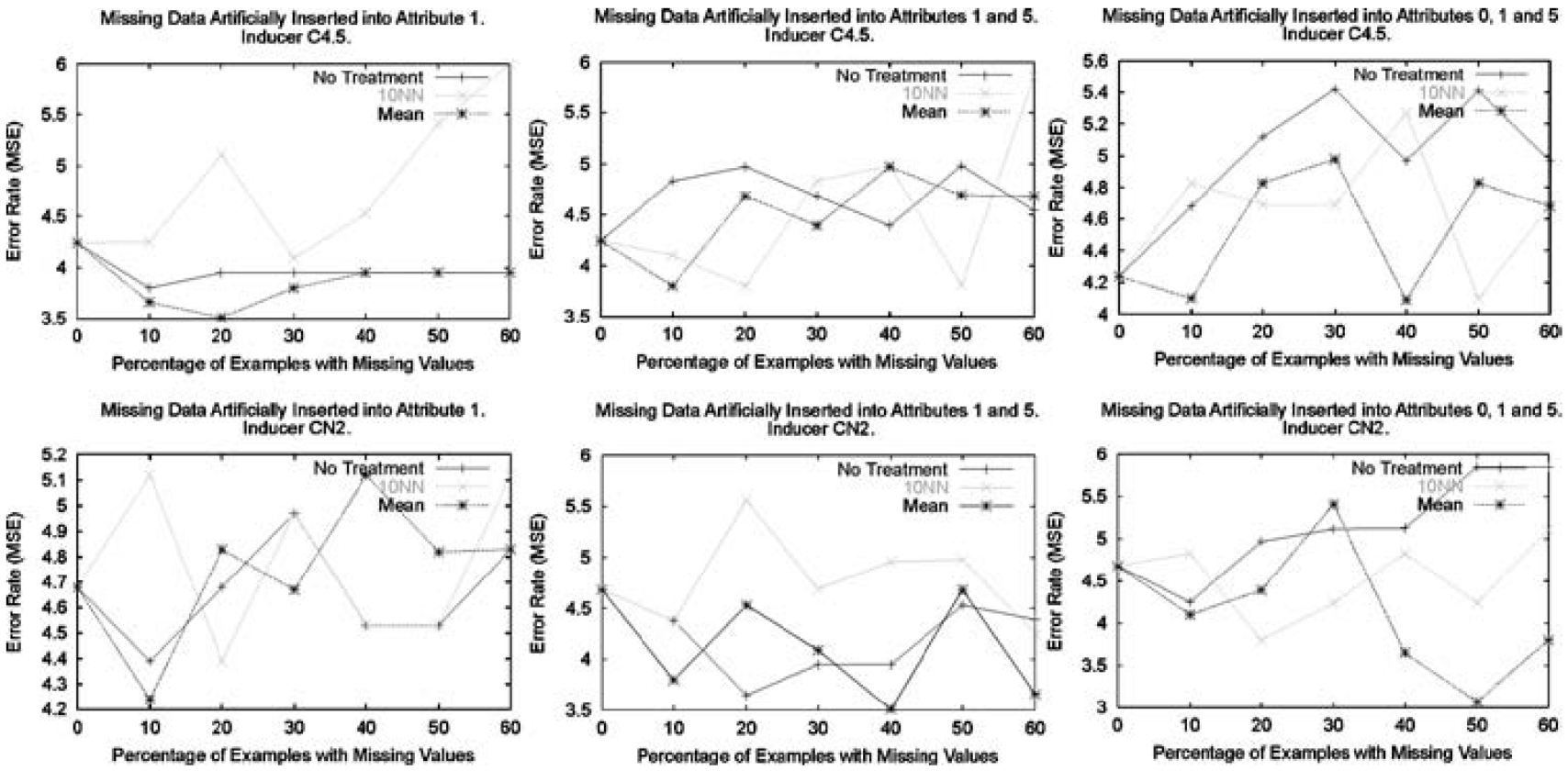
- Incomplete observations:

  $x_j = NA$
- Missingness mechanisms:[34]
- MCAR — Missing Completely At Random
  - Missingness independent of both observed and unobserved variables.
  - unbiased estimation possible, rare in real datasets.
- MAR — Missing At Random
  - Missingness depends only on observed variables.
    - Income missing more often for younger users.
- MNAR — Missing Not At Random
  - Missingness depends on the unobserved value itself.
    - High-income individuals less likely to report salary.

- Simple Imputation (baseline)
  - mean / median imputation
  - constant value imputation:
  - simple:, underestimates variance.
- Statistical Imputation
  - regression imputation
- ML-Based Approaches
  - kNN imputation
  - matrix completion
  - autoencoder-based imputation
  - generative models (VAE, diffusion)

IMPACT OF MISSING IMPUTATION

- Different techniques→
- Different results[35]



NOISE

- Feature noise, missing data, and outliers primarily degrade the quality of the learning signal rather than the supervision itself.
- Reduced Signal-to-Noise Ratio
  - Useful signal becomes harder to detect
  - models require more data to learn stable patterns
  - increased variance in parameter estimation.

$$SNR = \frac{Var(signal)}{Var(noise)}$$

- Noise increases denominator → weaker learnable structure.
- Observation corruption leads to:
  - distorted correlations between variables
  - artificial patterns introduced by preprocessing
  - misleading causal interpretations.

LABEL NOISE

- Label corruption can be modeled as:

$$P(y \mid y^{true}) = T$$

- where T is the noise transition matrix:

$$T_{ij} = P(y = j \mid y^{true} = i)$$

- diagonal terms refers to the correct labeling probability
- off-diagonal terms are systematic mislabeling.

- Symmetric (random) noise

$$P(y \neq y^{true}) = \eta$$

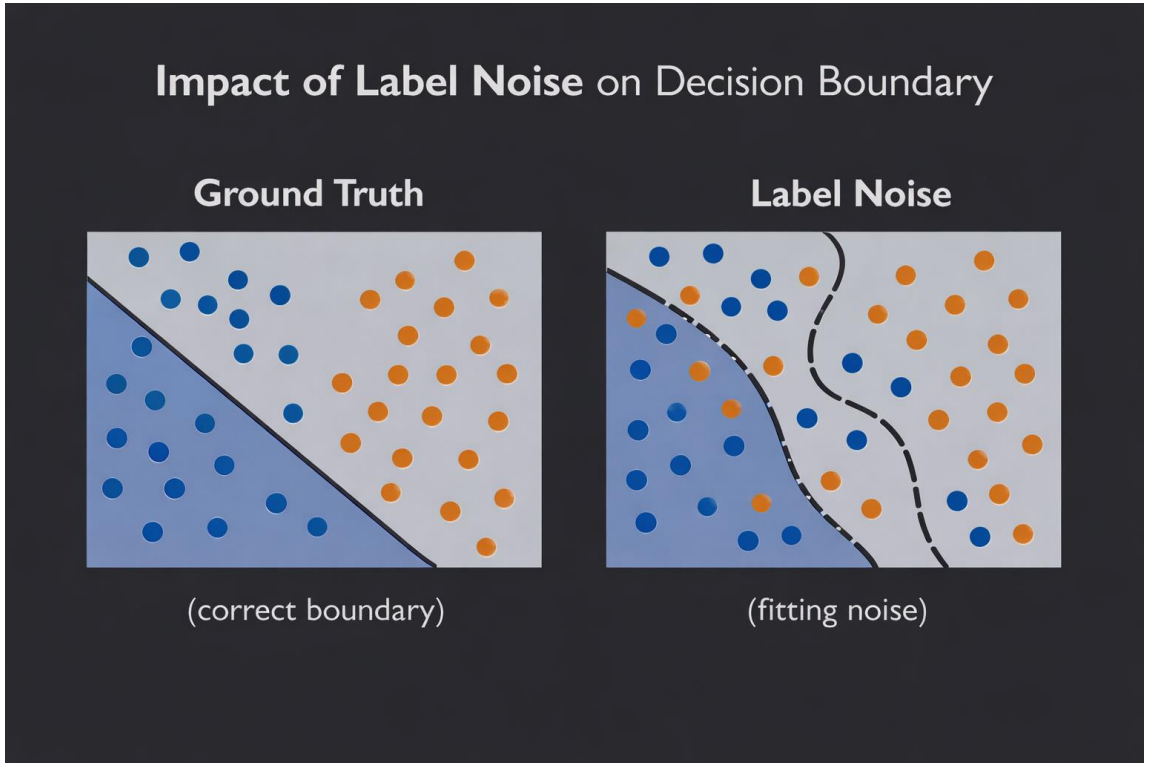
  - uniform mislabeling across classes.
- Asymmetric (class-dependent) noise
  - confusion between semantically similar classes
  - more realistic in real datasets.

LABEL NOISE

- Learning Impact[32]
  - Label noise alters empirical risk minimization:

$$\hat{R}(f) = \frac{1}{n}\sum_{i=1}^{n} \ell\left(f'(x_i)y_i\right)$$

  - since optimization uses corrupted supervision.
- Consequences:
  - biased decision boundaries
  - memorization by deep networks
  - degraded calibration.
- Label noise changes the teacher



FEATURE NOISE VS LABEL NOISE

Models tolerate feature noise better than label noise.

Feature noise

affects signal quality

often reducible

Label noise

affects supervision

often catastrophic

SAMPLING BIAS

- Sampling bias occurs when the training data distribution differs from the real-world population:

$$P_{train}(x) \neq P_{real}(x)$$

- The model learns patterns from an observed subset rather than from the true data-generating process.
- This mismatch leads to biased empirical risk minimization:

$$\hat{R}_{train}(f) \neq R_{real}(f)$$

- Possibile motivation
  - Data reflects past policies rather than reality.
    - Credit approvals trained only on previously accepted applicants.
  - Selective Observation
    - Data is collected only when events are observable.
    - Medical diagnoses only for tested patients
  - Survivorship Bias
    - Only successful or persistent entities remain in the dataset.
    - active companies vs failed startups

The model is not wrong
It is perfectly correct for the data it has seen.
self-reinforcing bias.

CLASS IMBALANCE

- Class imbalance occurs when class probabilities are highly skewed:

$$P(y_i) \ll P(y_j)$$

- where one or more classes are significantly underrepresented in the training data.
- Learning minimizes empirical risk:

$$\hat{R}(f) = \frac{1}{n}\sum_{i=1}^{n} \ell\,(f'(x_i)y_i)$$

- When classes are imbalanced:
  - majority class dominates the loss
  - minority errors contribute little to optimization.
- the model implicitly optimizes for the majority class.
- The accuracy paradox

- Class imbalance causes:
  - biased decision thresholds
  - poor minority recall
  - unstable probability calibration
  - misleading evaluation metrics.
- Data-Level
  - oversampling (SMOTE)
  - undersampling
  - synthetic data generation
- Algorithm-Level
  - class-weighted loss

$$\ell_w = w_y \cdot \ell(f'(x)y)$$

  - focal loss [36]
  - cost-sensitive learning.

DISTRIBUTION SHIFT

- Distribution shift occurs when the data distribution during deployment differs from the one observed during training:
- Covariate Shift

$$P_{train}(X) \neq P_{test}(X), P(Y \mid X) \textit{ unchanged}$$

- Inputs change but the underlying relationship remains stable.
  - user demographics evolve
  - sensor environments change.
- Mitigation:
  - importance weighting
  - domain adaptation.

- Label Shift (Prior Probability Shift)
  - Class proportions change.
- Example:
  - Fraud rate increases during economic crisis.
- Mitigation:
  - prior reweighting
  - calibration methods.

DRIFT

- Concept drift[37]
- The relationship between features and labels changes.
- Example:
  - fraud strategies evolve
  - language meaning changes over time.
- Most challenging type.

- Real-World Examples
  - Financial markets regime changes
  - Medical treatment evolution
  - User behavior adaptation
  - Recommendation systems feedback loops.
- Distribution shift causes:
  - performance degradation after deployment
  - model calibration failure
  - fairness instability
  - need for continuous monitoring.

Training assumes the future looks like the past.
Distribution shift is when this assumption stops being true.

DATA LEAKAGE

- Data leakage occurs when information unavailable at prediction time is unintentionally used during training.
- Target derived features
  - The model learns shortcuts that will not exist in real-world usage.
  - Features constructed using the target variable.
    - aggregated labels
    - future-derived indicators.
    - predict loan default using "number of missed payments after approval".
- Improper Normalization / Preprocessing
  - scaling using train + test data
  - feature selection before data split
  - PCA fitted globally.

- Temporal Leakage
  - Future information accidentally used for past predictions.
  - Predicting stock movement using indicators computed with future prices.
- Leakage leads to:
  - overly optimistic evaluation
  - non-reproducible results
  - deployment failure
  - incorrect scientific conclusions.

Leakage is the easiest way to publish a state-of-the-art model that completely fails in production.

TWO DIFFERENT NOTIONS OF DATA QUALITY

- From a Data Management Perspective
- correctness of stored data.
- Typical dimensions:
  - Completeness — required values are present
  - Consistency — no conflicting information
  - Accuracy— the distance between real value v and stored value v'
  - Schema correctness — valid formats and types

- *"Is the data correct?"*

- From a Learning perspective
- usefulness for learning predictive patterns.
- Some dimensions:
  - Predictive usefulness — features contain signal
  - Label reliability — supervision is trustworthy
  - Representativeness — dataset reflects deployment population
  - Stability under perturbation — robustness to small changes.

  - *"Is the data informative for prediction?"*

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